



Energy Consumption Driver

Final Report

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1. Introduction

1.1. Project Objectives

The energy consumption meter project was designed to achieve the following objectives:

- **Precise energy consumption measurement** – enabling accurate measurement and comparison of energy consumption by various electronic circuits.
- **Current pulse registration** – identification of excessive energy consumption moments and determination of minimum current efficiency of power sources.
- **Data analysis and archiving** – ability to save and read data from memory, allowing for results comparison and tracking changes over time.
- **Measurement documentation** – data transmission to a computer and presentation in a readable graphical form, facilitating report preparation.
- **Intuitive operation** – increasing laboratory technician efficiency through a user-friendly interface and automatic result saving.

1.2. System Description

The energy consumption driver is an advanced measurement device dedicated to analyzing energy consumption by electronic circuits, particularly STM32 family microcontrollers powered by 3.3 V. The system was designed to support engineers in the process of energy optimization of designed solutions.

The basis of the meter operation is the STM Nucleo F334R8 microcontroller, which manages the measurement process and user communication. Current pulse measurement is realized by monitoring the voltage drop across a measurement resistor connected in series with the tested circuit, using precision measurement amplifier (Current Sense Amplifier) and ADC converters operating at a minimum frequency of 1 Msps.

A unique feature of the system is the use of a supercapacitor as the power source for the tested circuit. This element, charged to 3.3 V, allows for realistic simulation of battery-powered devices. Continuous monitoring of the supercapacitor voltage enables determination of total energy consumption and instantaneous power based on discharge time.

The system offers modular construction with the ability to select different measurement and switching circuit variants using configuration jumpers. Measurement results are presented in real-time on an LCD display (supercapacitor voltage, percentage charge state, remaining time) and can be transmitted via UART interface to a PC, where a LabVIEW application enables detailed analysis and visualization in graph form.

1.3. Team Role Division

Within the project implementation, a division of responsibilities was applied ensuring effective collaboration utilizing diverse team member competencies:

- **Michał Dobrzycki** – project manager and systems integrator. Responsible for team work coordination, schedule oversight, and technical documentation preparation. Implementation of visualization application in LabVIEW environment, system requirements analysis, and hardware-software layer integration. User interface design and ensuring coherent operation of the entire measurement system.
- **Bartłomiej Osiak** – hardware design engineer. Design of electrical schematics and PCB printed circuit boards in KiCad environment. Selection of electronic components, connection correctness verification, and prototype production process oversight. Participate in LabVIEW application development.
- **Bartosz Hadała** – embedded software engineer. Firmware implementation for STM32F334R8 microcontroller in C language using HAL library. Peripheral programming, code optimization, and conducting functional tests and system validation. Collaboration in LabVIEW application development and participation in final project documentation preparation.

1.4. Project Timeline

The project was implemented according to the established timeline, covering key stages of system development – from the conceptual phase, through design and implementation, to testing and documentation.

Project Stage	Deadline
Function definition and requirements analysis	12.11.2025
System block diagram development	18.11.2025
PCB design in KiCad environment	21.11.2025
Printed circuit board manufacturing and assembly	30.11.2025
LabVIEW visualization application development	20.12.2025
Microcontroller software implementation	31.12.2025
Functional testing and system validation	06.01.2026
Technical documentation preparation	30.01.2026
Final project results presentation	30.01.2026

The timeline was developed taking into account dependencies between individual stages and ensuring adequate time for testing and possible corrections. Project implementation proceeded according to plan, enabling timely completion of all planned tasks.

2. Requirements Specification

2.1. Functional Requirements

The energy consumption meter system was designed with the following functionalities:

2.1.1. Measurements and Data Acquisition

- **Current measurement** – measurement of voltage drop across measurement resistor using precision amplifiers and ADC converter with minimum sampling frequency of 1 Msps
- **Supercapacitor voltage monitoring** – current tracking of power source charge level in the range of 2.5–3.3 V
- **Power peak analyze** – automatic detection and recording of short-term current consumption increases
- **Energy parameter determination** – calculation of instantaneous power, total energy consumed, and operating time based on supercapacitor discharge

2.1.2. User Interface

- **LCD display** – real-time presentation of supercapacitor voltage, percentage charge state, and time to go
- **LED signaling** – visualization of supercapacitor charging circuit operating state
- **Button control** – measurement initiation and termination, menu navigation

2.1.3. Communication and Archiving

- **UART data transmission** – transmission of measurement results to PC in real-time
- **Internal memory** – measurement history storage enabling later analysis and results comparison
- **LabVIEW integration** – data visualization in time graph form, power peak analysis, report export

2.1.4. Hardware Configuration

- **Measurement path selection** – ability to switch between different amplifiers (Current Sense Amplifier, operational amplifier) using jumpers
- **Anti-aliasing filter** – optional activation of high-frequency noise filtering
- **Power switch selection** – configuration of element type connecting supercapacitor with tested circuit (PMOS, relay, integrated circuit)

2.2. Non-functional Requirements

2.2.1. Performance and Reliability

- **Operational stability** – continuous error-free operation during long measurement sessions
- **Circuit redundancy** – alternative measurement paths ensuring work continuation in case of component failure
- **Power management** – microcontroller power consumption optimization to minimize measurement impact

2.2.3. Usability

- **Intuitive operation** – simple interface requiring no advanced technical knowledge
- **Display readability** – well-visible information in various lighting conditions
- **Quick configuration** – ability to change measurement components without software reload necessity

2.2.4. Communication Interfaces

- **UART/USB** – stable data transmission with error handling capability
- **Communication protocol** – standardized data format facilitating integration with various programming environments
- **Compatibility** – cooperation with LabVIEW application on Windows/Linux systems

2.2.5. Hardware Requirements

- **Compact construction** – PCB dimensions enabling easy integration in laboratory environment
- **Modularity** – ability to replace and configure components without entire circuit re-design necessity
- **Manufacturing quality** – professional soldering, appropriate insulation and shielding for interference elimination

3. System Design

3.1. System Architecture

3.3.1. Measurement Section

Section responsible for precise electrical parameter monitoring:

- **Two measurement amplifiers** measuring voltage drop across measurement resistor:
 - Current Sense Amplifier – dedicated current measurement amplifier
 - Measurement amplifier – alternative measurement circuit
- **Anti-aliasing filter** at amplifier output – user has the ability to enable and disable in variant selection section, allowing elimination of high-frequency noise.

All measurement variants are selected by the user by shorting appropriate pins in the control section. Voltage measurement across the resistor is performed by ADC1 converter in the microcontroller.

3.3.2. Switching Section

Responsible for controlled connection of supercapacitor with tested circuit. Three switch variants are available:

- **PMOS transistor** – semiconductor switching element
- **Reed relay switch** – mechanical switching element

- **Dedicated integrated circuit** – specialized chip for power management

The user has complete freedom in switching method selection. All switching circuits are controlled using microcontroller GPIO pins.

3.3.3. Power Supply and Charging Section

Manages supercapacitor charging and monitoring process:

- **Integrated “charger” circuit** – dedicated chip performing intelligent supercapacitor charger role, controlled via GPIO.
- **Supercapacitor** – main power source for tested circuit, remaining under constant ADC2 microcontroller converter supervision.

3.3.4. Results Display Section

User interface ensuring measurement visualization:

- **LCD screen 2×16 characters** – basic device-human communication interface, displaying:
 - Current supercapacitor voltage value (data from ADC2)
 - Percentage supercapacitor charge value, normalized to range:
 - * 2.5 V → 0%
 - * 3.3 V → 100%
 - Estimated measurement time
- **Two LED diodes** – signaling of supercapacitor charging circuit operating state.

3.3.5. Configuration Section

Jumper system with functionality labels, enabling:

- Measurement amplifier variant selection
- Anti-aliasing filter enable/disable
- Power switch type selection

4. Implementation

4.1. Tools and Technologies

4.1.1. Hardware

- **Microcontroller:** STM Nucleo F334R8
- **Display:** LCD 2×16 characters
- **Energy storage element:** 25 F Supercapacitor
- **Measurement circuits:** Current Sense Amplifier, operational amplifiers
- **Switching elements:** PMOS transistor, reed relay switch, dedicated integrated circuits

4.1.2. Software

- **MCU development environment:** STM32CubeIDE
- **Data visualization:** LabVIEW
- **Communication:** UART/USB
- **Version control:** Git

4.2. PCB Design

The printed circuit board was designed in KiCad environment. Below are detailed circuit schematics illustrating individual functional sections of the system and PCB connection routing.

Below are detailed PCB electrical circuit schematics illustrating individual functional sections of the system.

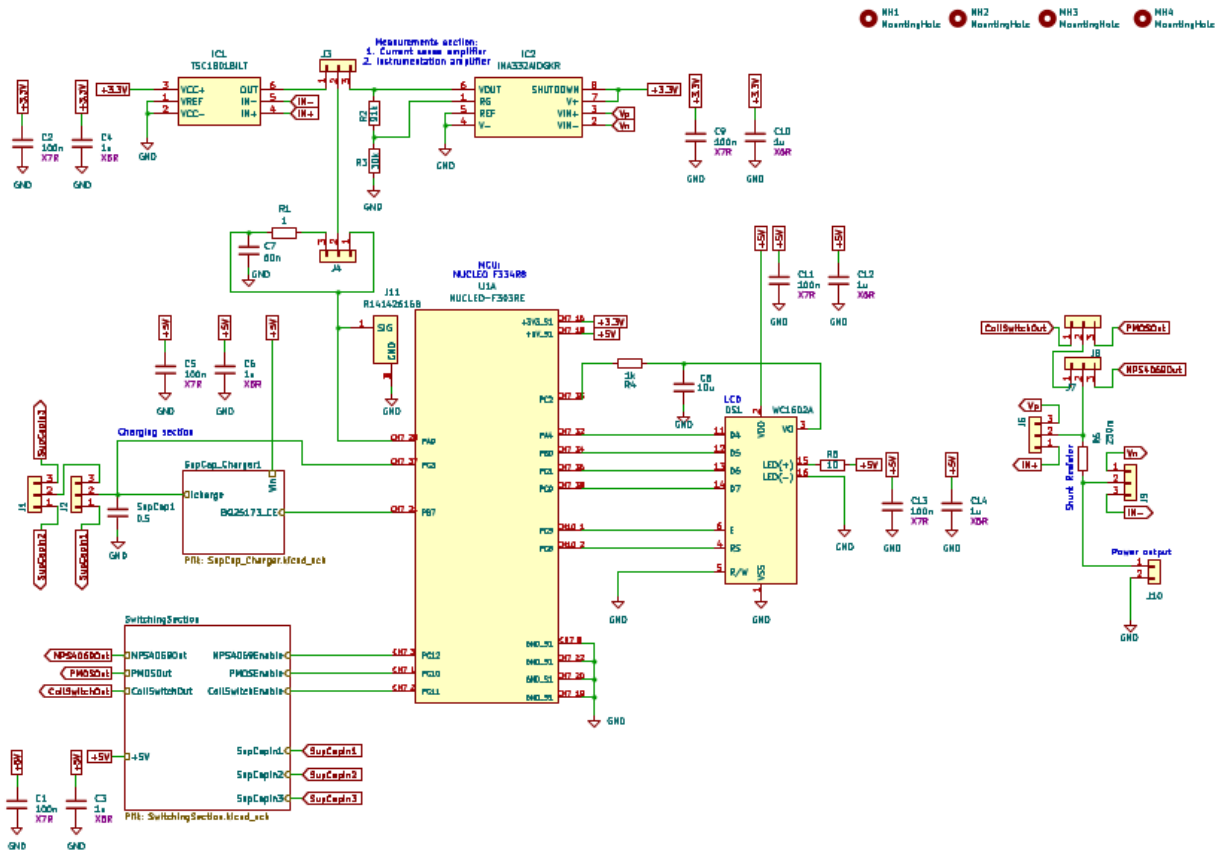
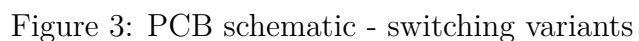


Figure 1: PCB schematic - main view

4.2.2. PCB Layout

Below is a view of the printed circuit board copper layers, illustrating trace routing and component placement.



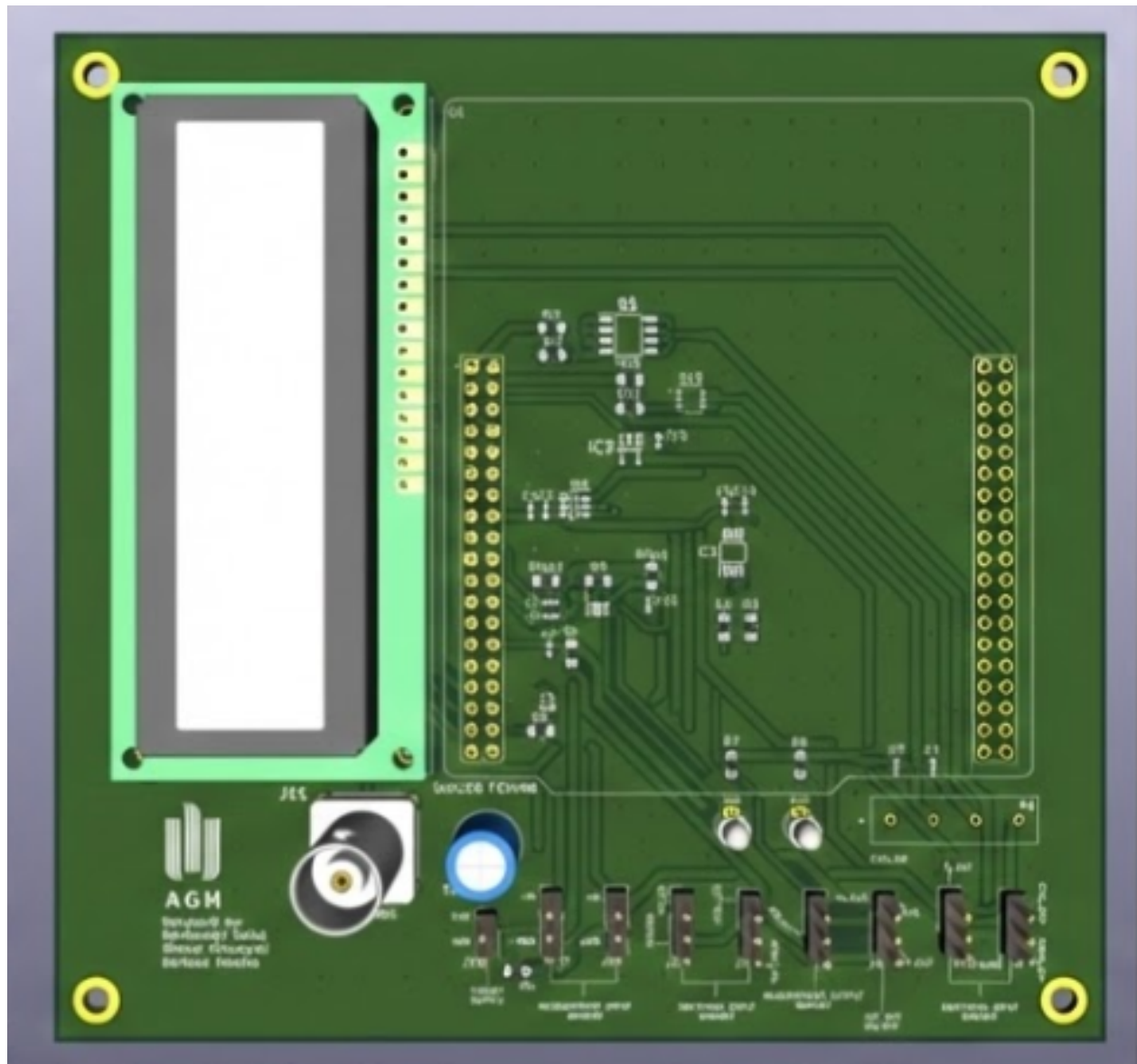


Figure 4: 3D visualization - front

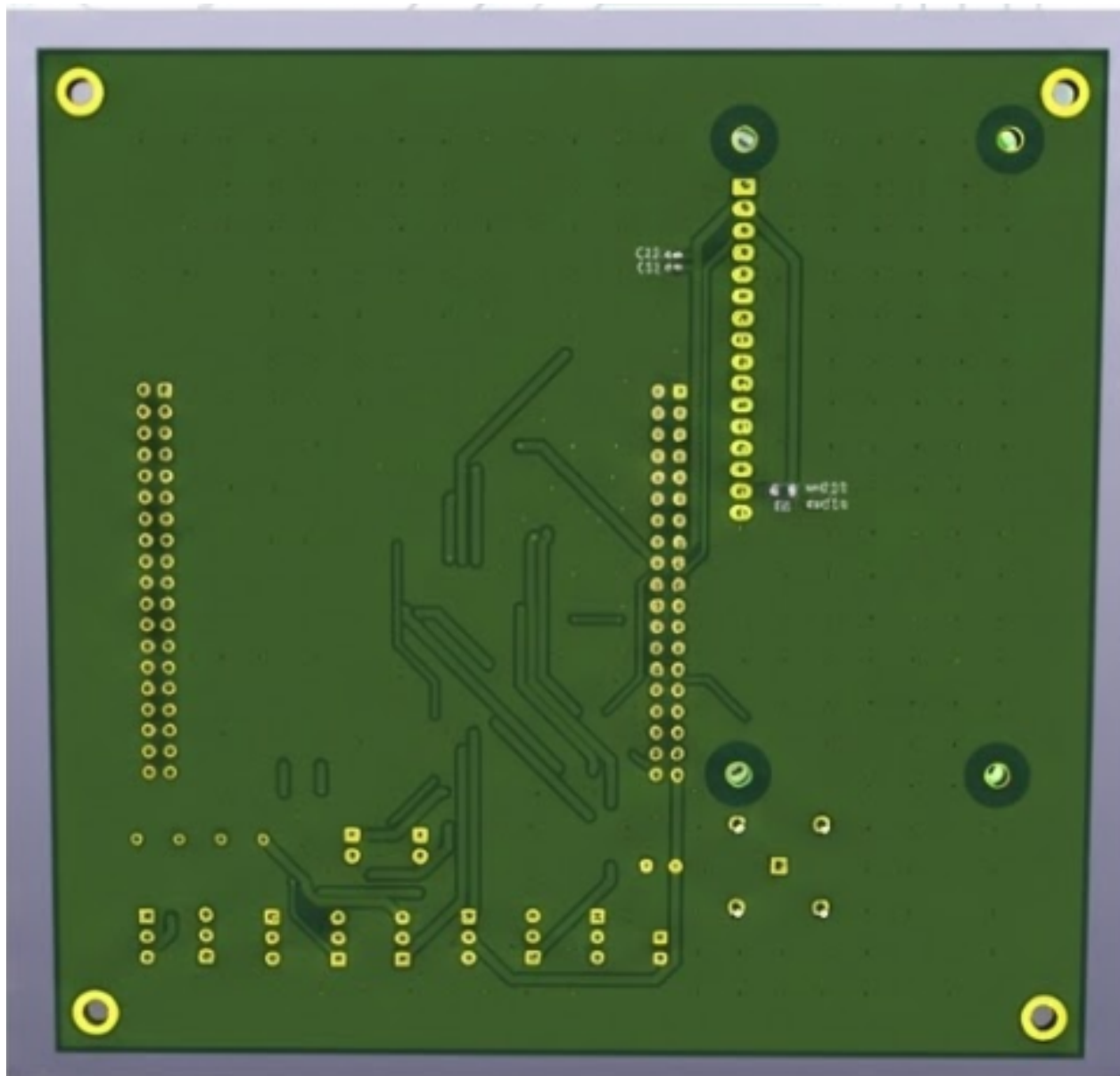


Figure 5: 3D visualization - back

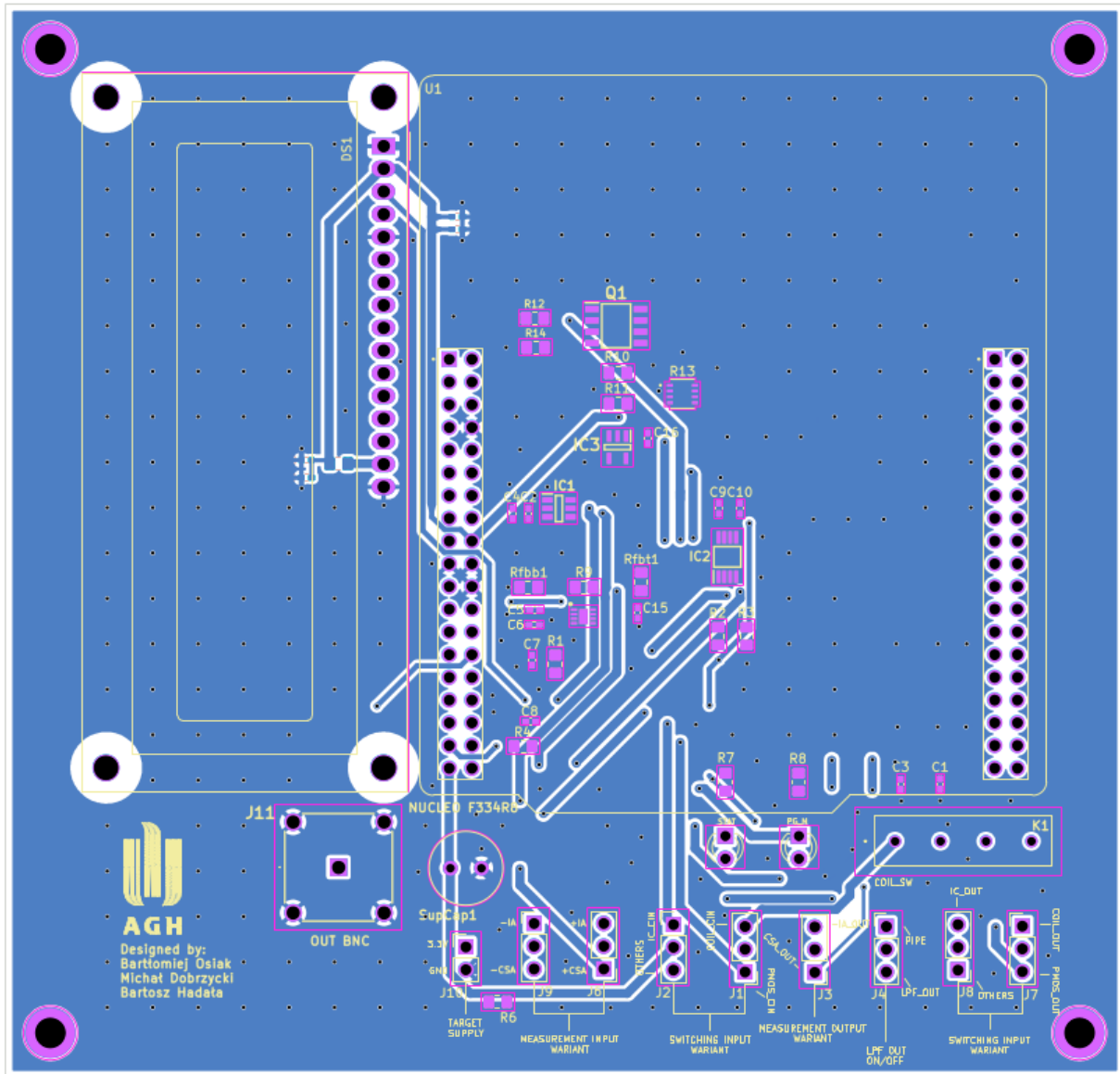


Figure 6: PCB layout - bottom layer view

4.3. Device Prototype

Below are photos of the manufactured energy consumption meter prototype. The designed and manufactured PCB contains all necessary functional sections: measurement, switching, power supply, and user interface.

The completed device constitutes a fully functional measurement system, enabling precise analysis of energy consumption by electronic circuits.

4.4. Microcontroller Software Implementation

4.4.1. Firmware Architecture

The STM32F334R8 microcontroller software was implemented in C language using the HAL (Hardware Abstraction Layer) library. The firmware architecture is based on a modular approach that ensures code readability and ease of functionality expansion.

Main application modules:

- **Measurement module** – ADC converter handling with 1 Msps sampling frequency, utilizing interrupts for data transfer without processor load
- **Communication module** – UART protocol implementation for data transmission to PC and LabVIEW application
- **Display module** – library created by Bartłomiej Osiak, LCD display control, presentation of voltage, charge state, and stored energy
- **Power management module** – supercapacitor charging control, charge level monitoring
- **Switching module** – electronic switch control, measurement variant selection

Firmware utilizes advanced real-time system mechanisms: - Interrupt handling for fast event response - State machines for operating mode management - Circular buffers for efficient measurement data management - Watchdog timer for increased system reliability

4.5. LabVIEW Visualization Application

The system was equipped with a visualization application created in LabVIEW environment, which receives data from the meter via UART interface and presents it in clear graphical form.

4.5.1. System Calibration

After system fabrication, it is necessary to perform a calibration procedure that ensures reading accuracy.

4.5.2. Main Visualization Panel

The main LabVIEW application panel enables real-time monitoring of key energy parameters and presents graphs of energy consumption and instantaneous power.

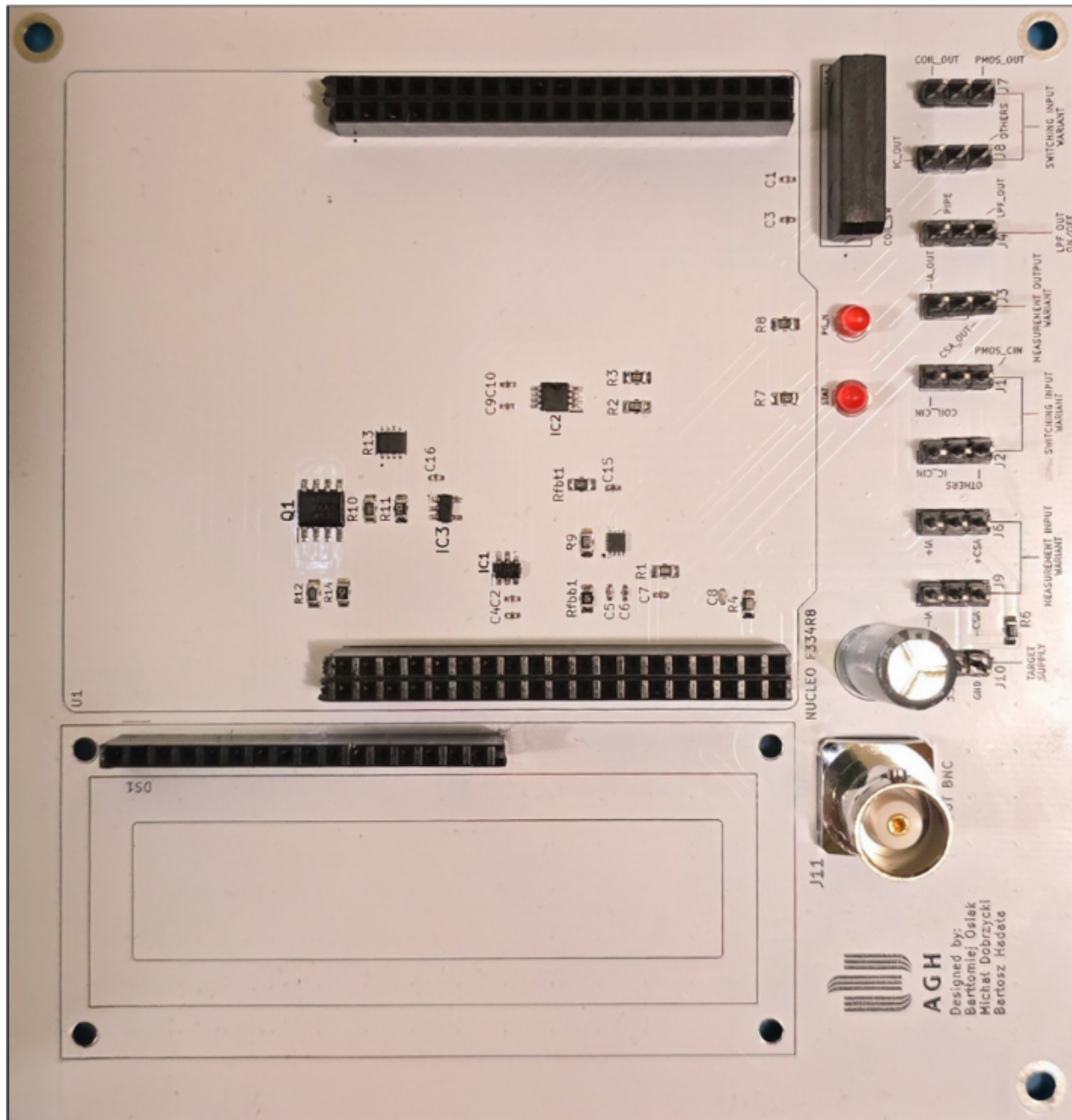


Figure 7: Meter prototype - view 1

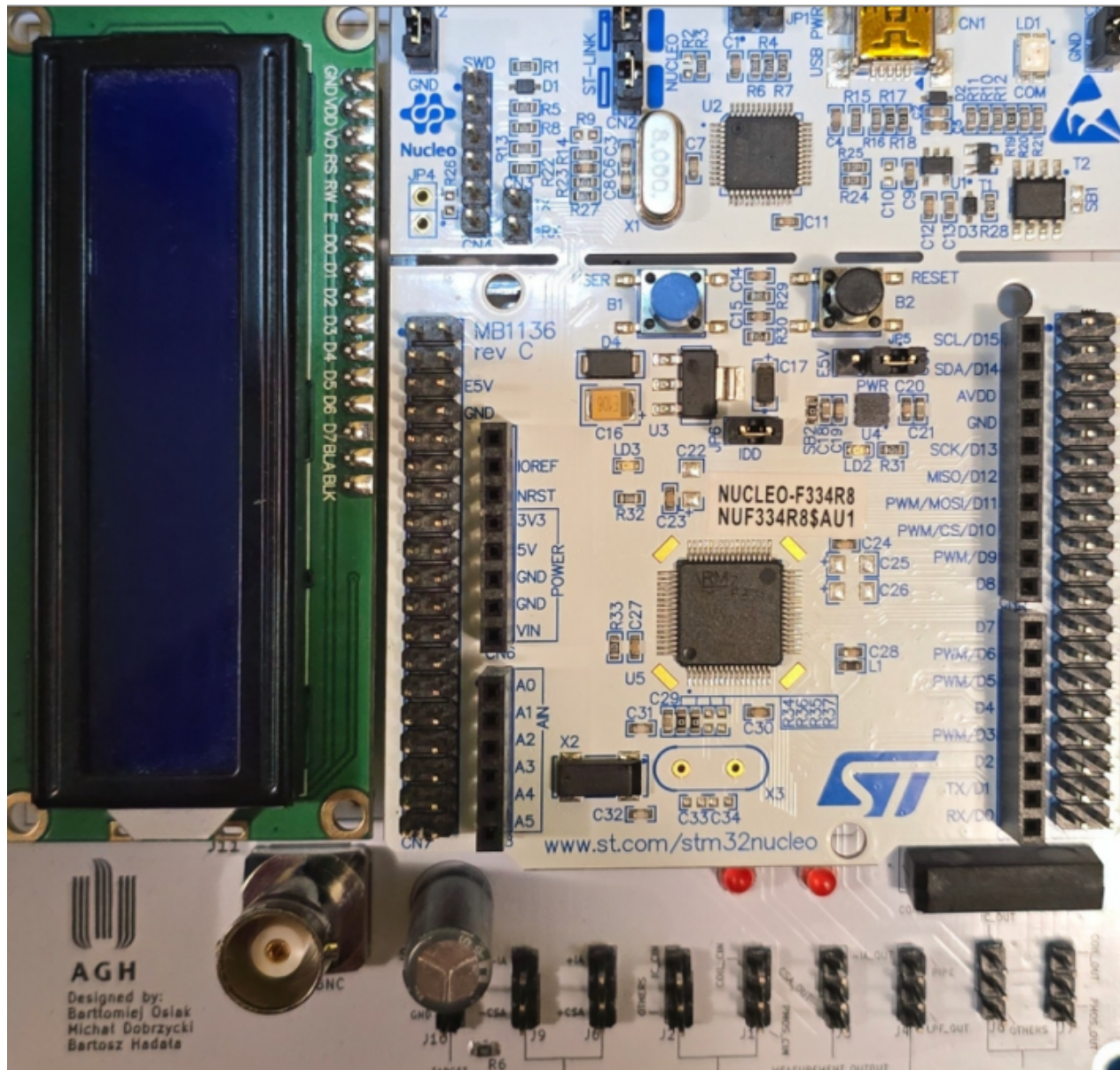


Figure 8: Meter prototype - view 2

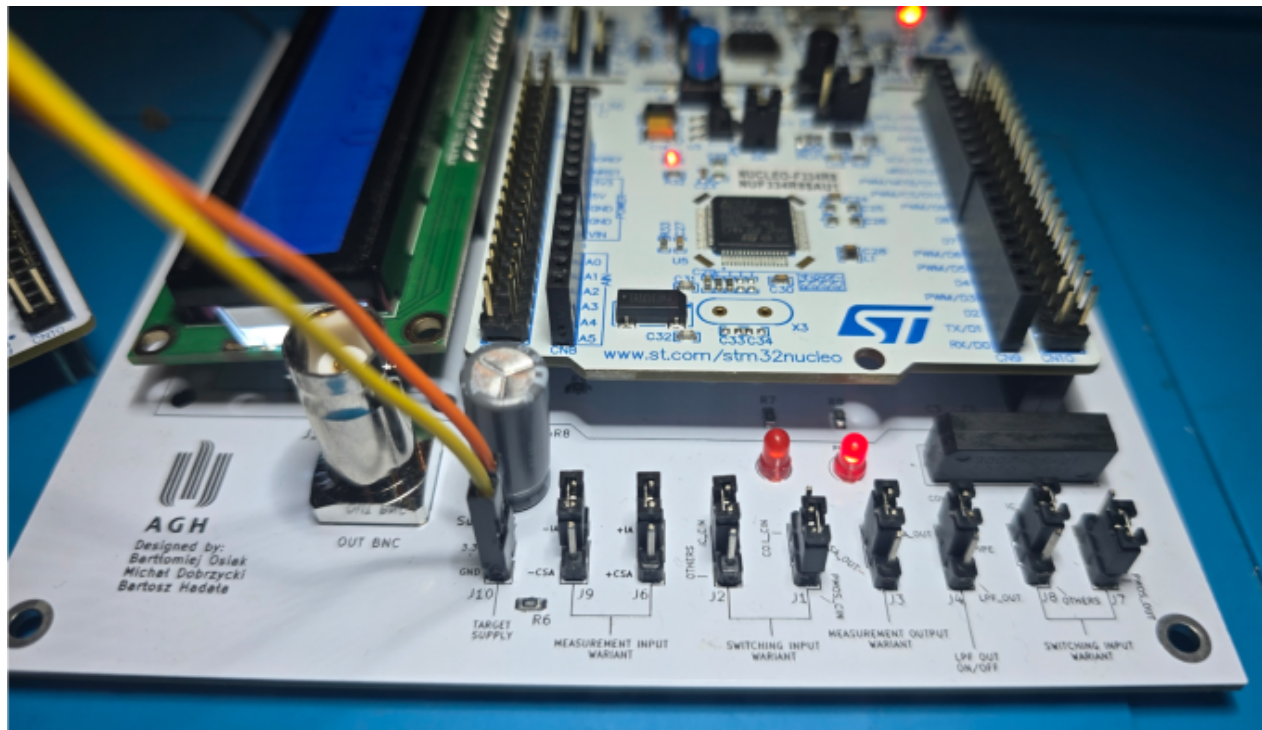


Figure 9: Meter prototype - view 3

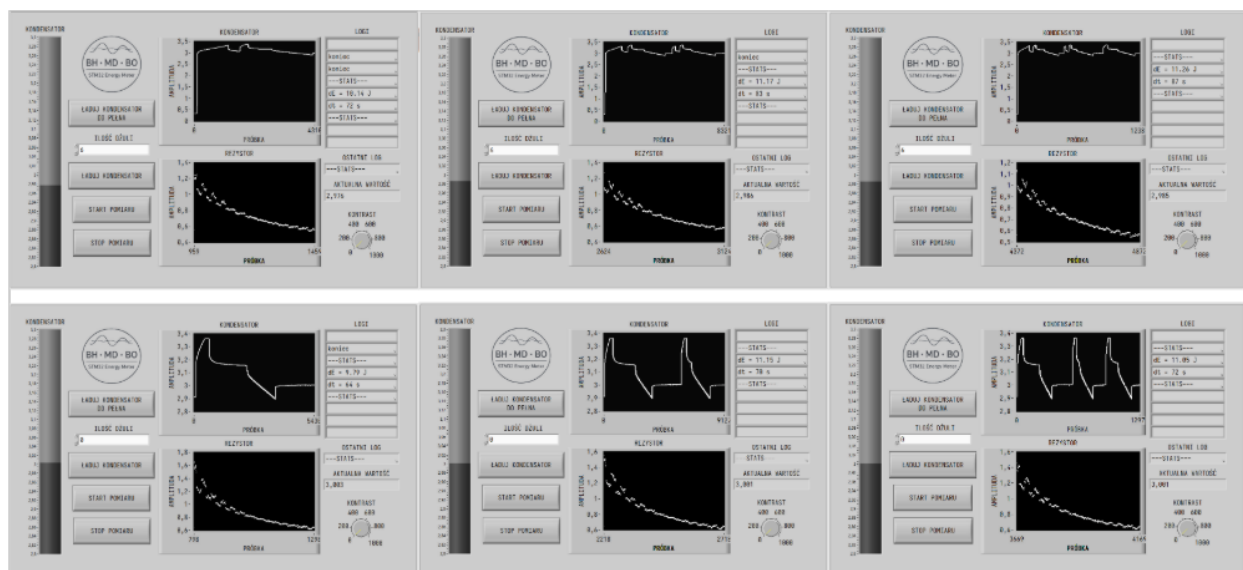


Figure 10: Calibration in LabVIEW

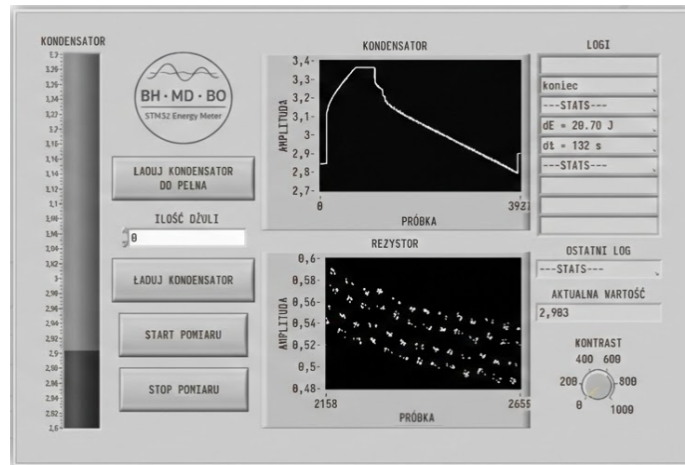


Figure 11: Main LabVIEW visualization interface

The application allows for: - Real-time voltage and current monitoring - Time graph visualization of power consumption - Measurement data recording and saving - Stored energy and operating time analysis - Power pulse detection and presentation

5. Testing and Validation

5.1. Hardware Testing

After soldering completion, detailed verification of component connection quality on the PCB was conducted. Particular attention was paid to possible cold solder joints and possible component overheating. For this purpose, connection continuity and correctness were thoroughly checked using a multimeter and microscope.

5.1.1. Detected Problems

During testing, one significant problem was encountered – the measurement amplifier in the sampling and amplification section did not work correctly. The cause was insufficient supply voltage.

Fortunately, the designed PCB contains redundancy in the form of an alternative measurement path variant. Thanks to the jumper system enabling selection of different configurations, it was possible to switch the circuit to the working Current Sense Amplifier, which allowed work continuation without hardware modifications necessity.

5.2. Software Testing and Optimization

5.2.1. User Interface Improvements

As part of software development, the following visual and functional corrections were introduced:

- **Log tables** – an event logging system was added enabling tracking of measurement history and system events.
- **Energy parameter display** – information presented on the LCD screen was expanded to include additional parameters such as stored energy, current power, and estimated time remaining until supercapacitor discharge.

5.2.2. Code Refactoring

At the end of the programming stage, complete refactoring of backend code running on the STM32F334R8 microcontroller was performed. Main refactoring goals included:

- **Architecture modularization** – code division into clearly separated modules responsible for specific functionalities (measurements, communication, display, power management).
- **Flexibility and extensibility** – application of design patterns enabling easy addition of new functionalities and modification of existing ones without base code intervention necessity.
- **Readability and maintainability** – naming standardization, addition of comments and function documentation, facilitating further development and code maintenance.

Thanks to the conducted refactoring, the application became more stable, easier to maintain, and prepared for future functionality expansions.

6. Summary and Conclusions

The “Energy Consumption Meter” project was implemented according to the established schedule and meets all set objectives. The developed device constitutes a comprehensive tool for precise measurement and analysis of energy consumption by electronic circuits, with particular emphasis on STM32 family microcontrollers.

6.1. Achieved Objectives

All main project objectives were achieved:

1. **Accurate energy consumption measurement** – the system enables precise current consumption monitoring with minimum sampling frequency of 1 Msps.
2. **Power pulse registration** – the device effectively detects and records moments of excessive energy consumption.
3. **Results analysis and comparison** – built-in memory and LabVIEW application allow for detailed analysis of collected data.
4. **Measurement documentation** – ability to export data to computer and generate reports facilitates documentation preparation.
5. **Intuitive interface** – LCD display and LabVIEW application ensure easy operation and quick access to information.

6.2. Innovative Solutions

The project is distinguished by several significant features:

- **Construction modularity** – ability to select different measurement and switching circuit variants ensures flexibility and failure resistance.
- **Modern firmware architecture** – utilization of advanced programming techniques (interrupts, state machines) guarantees performance and stability.
- **Comprehensive visualization system** – LabVIEW application offers advanced data analysis and presentation capabilities.
- **Supercapacitor as power source** – enables realistic simulation of battery-powered device operation.

6.3. Unrealized Objectives

Despite successful implementation of most set project objectives, due to time and technical constraints, full measurement data archiving functionality could not be implemented. The system in its current form enables real-time data transmission to LabVIEW application and display on LCD, however lacks a mechanism for long-term measurement storage directly in device memory. Planned archiving functionality included Flash memory measurement data storage, historical measurement session reading capability, collected data export to PC, and automatic report creation from conducted measurements. Implementation of this functionality constitutes a natural direction for further system development and can be realized in future project iterations.

The developed energy consumption meter constitutes a valuable tool for engineers involved in designing energy-efficient electronic systems and can find application in both academic and industrial environments.

6.4. Acknowledgments

The project authors wish to express sincere gratitude to all persons who contributed to the successful implementation of this project:

- **Jacek Kołodziej, PhD, DSc** – for valuable substantive guidance, coordination, and scientific support during project implementation
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